Reliability Engineering 101 for NSWC Corona

Jordan Hanson

July 20, 2026

Whittier College Department of Physics and Astronomy

Outline

Part I : Introduction and Mathematics

Outline

Part II : Reliability of Components and Systems

- 2.1 : Reliability of Components
    - 2.1.1 : Reliability Functions : $f(t)$, $F(t)$, $R(t)$, and $h(t)$
    - 2.1.2 : Common PDFs for Reliability Functions
    - 2.1.3 : Model Selection and Parameter Estimation
        - Application of Maximum-Likelihood Estimators (MLEs)
        - Non-parametric Estimators
        - Bayesian Techniques
- 2.2 : Reliability of Systems
    - 2.2.1 : Reliability Block Diagrams (RBDs)
    - 2.2.2 : Success and Fault Tree Analysis
    - 2.2.3 : Failure Mode and Effect Analysis (FMEA/FMECA)

Outline

Part III : Reliability and Availability of Repairable Systems

- 3.1 : Reliability of Repairable Systems
  - 3.1.1 : Point Process Definition and Homogeneous Point Processes (HPPs)
  - 3.1.2 : Non-Homogeneous Point Processes (NHPPs) and Renewal Processes
  - 3.1.3 : Analysis Techniques for HPPs and NHPPs
  - 3.1.4 : Availability of Repairable Systems
  - 3.1.5 : Availability of Complex Systems
- 3.2 : Selected Topics in Reliability Data Analysis
  - 3.2.1 : System Classification (One-shot, repairable)
  - 3.2.2 : Reliability Data, Type and Classification

Part I: Introduction and Mathematics

Module 1.2 : Mathematics of Reliability Engineering

Event Probability Vocabulary

- Event probability : $P(A)$
- Intersection of events : $P(A \cdot B) = P(A) \cdot P(B|A)$
- Union of events : $P(A + B) = P(A) + P(B) - P(A)P(B|A)$
- Independent events :
    - $P(A|B) = P(A)$
    - $P(B|A) = P(B)$
    - $P(A \cdot B) = P(A) \cdot P(B)$
    - $P(A + B) = P(A) + P(B) - P(A)P(B)$
- Mutually exclusive events :
    - $P(A \cdot B) = 0$
    - $P(A + B) = P(A) + P(B)$

Examples of event probability

- Suppose that Vendor 1 provides 40% and Vendor 2 provides 60% of the circuit boards used in a computer. It is further known that 2.5% of Vendor 1's supplies are defective and only 1% of Vendor 2's supplies are defective.

  Question : (i) What is the probability that a unit is both defective and supplied by Vendor 1? (ii) What is the same probability for Vendor 2?

  Define the events:

  - $E_1 =$ Circuit board is from Vendor 1
  - $E_2 =$ Circuit board is from Vendor 2
  - $D =$ Circuit board is defective

  Hence,

  - $\text{Pr}(E_1) = 0.40$
  - $\text{Pr}(E_2) = 0.60$
  - $\text{Pr}(D|E_1) = 0.025$
  - $\text{Pr}(D|E_2) = 0.010$

Examples of event probability

- Suppose that Vendor 1 provides 40% and Vendor 2 provides 60% of the circuit boards used in a computer. It is further known that 2.5% of Vendor 1's supplies are defective and only 1% of Vendor 2's supplies are defective.

  Question : (i) What is the probability that a unit is both defective and supplied by Vendor 1? (ii) What is the same probability for Vendor 2?

  Calculate probabilities:

  - (i) Probability that a unit is defective and supplied by Vendor 1: $P(E_1 \cdot D) = P(E_1) \cdot P(D|E_1) = 0.4 \times 0.025 = 0.01$
  - (ii) Probability that a unit is defective and supplied by Vendor 2: $P(E_2 \cdot D) = P(E_2) \cdot P(D|E_2) = 0.6 \times 0.01 = 0.006$

Examples of event probability

- Suppose that Vendor 1 provides 40% and Vendor 2 provides 60% of the circuit boards used in a computer. It is further known that 2.5% of Vendor 1's supplies are defective and only 1% of Vendor 2's supplies are defective.

  Question : (iii) What is the probability that a unit is defective, regardless of the vendor?

  Define the events:

  - $D =$ Defective board from vendor 1, **OR** defective board from vendor 2.
  - The sub-events are *mutually exclusive*, so $P(D) = P(D|E_1) + P(D|E_2)$

  Calculate probabilities:

  - $\Pr(D) = 0.01 + 0.006 = 1.6\%$

Examples of event probability

- Suppose that Vendor 1 provides 40% and Vendor 2 provides 60% of the circuit boards used in a computer. It is further known that 2.5% of Vendor 1's supplies are defective and only 1% of Vendor 2's supplies are defective.

  Question: (iv) What is the probability that a randomly selected board is either supplied by Vendor 1 or is defective?

  Define the events:

  - $E_1 =$ Board supplied by Vendor 1.
  - $D =$ Board is defective.
  - We seek the probability for E_1 OR D, but these are *not mutually exclusive.*

  Calculate probabilities:

  - $P(E_1 + D) = P(E_1) + P(D) - P(D|E_1)$
  - $P(E_1 + D) = 0.4 + 0.016 - 0.01 = 40.6\%$

Examples of event probability

- A particular type of computer chip is manufactured by three different suppliers. It is known that 5% of chips from Supplier 1, 3% of Supplier 2, and 8% from Supplier 3 are defective.

  Question: (i) If one chip is selected from each supplier, what is the probability that one of the chips is defective? (ii) Two chips?

  Define the events:

  - $D_1 =$ Chip from Supplier 1 is defective.
  - $D_2 =$ Chip from Supplier 2 is defective.
  - $D_3 =$ Chip from Supplier 3 is defective.
  - Note that these events are independent.

  Formula for the probability of the union of two independent events:

  - $P(E_1 + E_2) = 1 - (1 - P(E_1))(1 - P(E_2))$
  - Expands to original formula, and also extends to N events.

Examples of event probability

- A particular type of computer chip is manufactured by three different suppliers. It is known that 5% of chips from Supplier 1, 3% of Supplier 2, and 8% from Supplier 3 are defective.

  Question: (i) If one chip is selected from each supplier, what is the probability that one of the chips is defective? (ii) Two chips?

  Calculate the probability for (i):

  - $P(E_1 + E_2 + E_3) = 1 - (1 - P(E_1))(1 - P(E_2))(1 - P(E_3))$
  - $P(E_1 + E_2 + E_3) = 15.2\%$

Examples of event probability

- A particular type of computer chip is manufactured by three different suppliers. It is known that 5% of chips from Supplier 1, 3% of Supplier 2, and 8% from Supplier 3 are defective.

  Question: (i) If one chip is selected from each supplier, what is the probability that one of the chips is defective? (ii) Two chips?

  Calculate the probability for (ii):

  - E_1 and E_2, not E_3 : $0.05 \times 0.03 \times (1 - 0.08) = 0.00138$
  - E_1 and E_3, not E_2 : $0.05 \times (1 - 0.03) \times 0.08 = 0.00388$
  - E_2 and E_3, not E_1 : $(1 - 0.05) \times 0.03 \times 0.08 = 0.00228$
  - Total probability: 0.754% (Notice this is most of the difference between 1-chip case and summing all failure probabilities to 16%).

Bayes' Theorem

- A general formula for the probability of events that are not necessarily independent, or mutually exclusive is **Bayes' Theorem.**

  Proof: Let A and B be two events. The intersection of A and B may be written:

  - $P(A \cdot B) = P(A) \cdot P(B|A)$
  - $P(A \cdot B) = P(B) \cdot P(A|B)$
  - It follows that $P(A) \cdot P(B|A) = P(B) \cdot P(A|B)$
  - Solve for $P(A|B)$:

$$\boxed{P(A|B) = P(A)P(B|A)/P(B)} \tag{1}$$

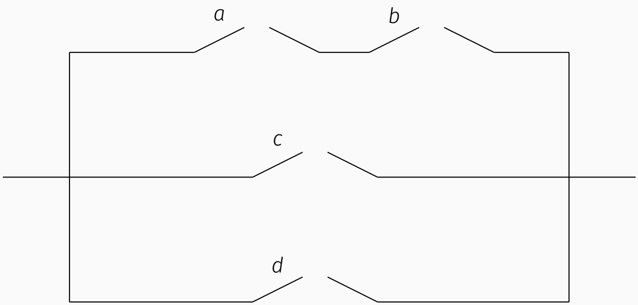
- **Generalization :** Let E be an event from a set of events A_n. The probability that A_j occurs, given E is

$$\boxed{P(A_j|E) = \frac{P(A_j)P(E|A_j)}{\sum_i^n P(A_i)P(E|A_i)}} \tag{2}$$

Bayes' Theorem

In the circuit shown, the probability of any switch being closed is 0.8, and all events are independent. (i) What is the probability that a circuit will exist? (ii) Given that a circuit exists, what is the probability that switches a and b are closed?

Define events : Let the events that switches a, b, c, and d are closed be denoted by A, B, C, and D, respectively. Let X denote the event that a complete circuit exists.



Bayes' Theorem

In the circuit shown, the probability of any switch being closed is 0.8, and all events are independent. (i) What is the probability that a circuit will exist? (ii) Given that a circuit exists, what is the probability that switches *a* and *b* are closed?

(i) The circuit exists if either the top branch is completed or at least one of the middle or bottom branches is completed. This means that $X = AB + (C + D)$. Take the probability of this expression, and use the fact that the events are independent:

$$P(X) = P(AB) + P(C + D) - P(AB)P(C + D) \tag{3}$$

$$P(AB) = P(A)P(B) = (0.8)(0.8) = 0.64 \tag{4}$$

$$P(C + D) = P(C) + P(D) - P(C)P(D) \tag{5}$$

$$= 0.8 + 0.8 - (0.8)(0.8) = 0.96 \tag{6}$$

Combine results:

$$P(X) = 0.64 + 0.96 - (0.64)(0.96) = 0.9856. \tag{7}$$

Bayes' Theorem

In the circuit shown, the probability of any switch being closed is 0.8, and all events are independent. (i) What is the probability that a circuit will exist? (ii) Given that a circuit exists, what is the probability that switches *a* and *b* are closed?

(ii) Using conditional probability,

$$P(AB|X) = \frac{P(ABX)}{P(X)}. \tag{8}$$

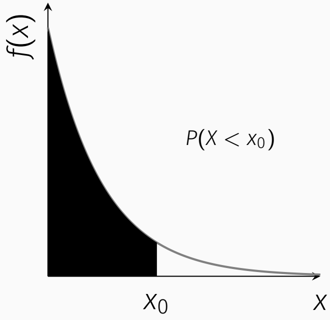
Since the occurrence of AB guarantees that a circuit exists, $ABX = AB$, and $P(ABX) = P(AB)$. Thus,

$$P(AB|X) = \frac{P(AB)}{P(X)} = \frac{P(A)P(B)}{P(X)} = \frac{(0.8)(0.8)}{0.9856} = 0.6494. \tag{9}$$

Probability distributions describe the *density* of probability.

- Discrete distributions
  - Binomial
  - Poisson
- Continuous distributions
  - Exponential
  - Gaussian
  - Weibull
  - Gamma and Beta

Right : The exponential distribution used to compute $P(X < x_0)$



$$P(X > x_0) = \int_{x_0}^{\infty} \lambda e^{-\lambda x} dx \quad (10)$$

Example : A pressure sensor has a constant failure rate

$$\lambda = 2 \times 10^{-5} \text{ failures/hour.} \tag{11}$$

Question : Assume the sensor lifetime is exponentially distributed. What is the probability that the pressure sensor operates without failure for at least 20,000 hours?

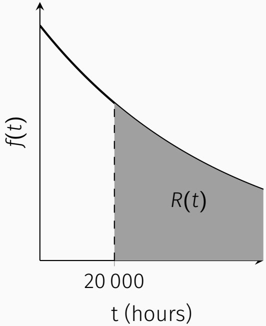
Show[a] that the exponential reliability function is

$$R(t) = P(T > t) = e^{-\lambda t}. \tag{12}$$

Substitute the given values:

$$\boxed{R(20000) = 0.670} \tag{13}$$

The probability that the pressure sensor survives at least 20,000 hours is 0.670 (67.0%).



[a] The PDF of time to failure is $f(t)$, so $R(t)$ is $1 - F(t)$, where $F(t)$ is the cumulative distribution function (CDF) of $f(t)$.

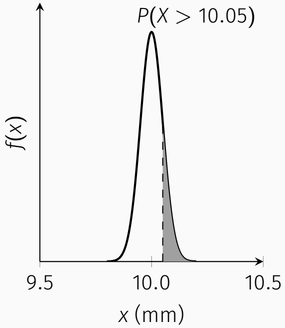
Example : The diameter of a manufactured pressure sensor diaphragm is normally distributed with $\mu = 10.00$ mm and $\sigma = 0.05$ mm.

Question : What is the probability that a randomly selected diaphragm has a diameter greater than 10.05 mm?

The Gaussian probability density function is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]. \qquad (14)$$

The corresponding z-score is $z = (x-\mu)/\sigma = (10.05 - 10.00)/0.05 = 1$. Using the standard normal table, $P(X > 10.05) = P(Z > 1) = 0.1587$. There is a 15.9% chance that the diameter exceeds 10.05 mm.



Any normal distribution can be converted to the **standard normal distribution**:

$$z = \frac{x - \mu}{\sigma} \tag{15}$$

The probability that a value lies below x is $P(X < x) = P(Z < z)$, and is found using a **z-table**.

Example: Standard Normal CDF Table

z	0.00	0.01	0.02	0.03
0.0	0.5000	0.5040	0.5080	0.5120
0.1	0.5398	0.5438	0.5478	0.5517
0.2	0.5793	0.5832	0.5871	0.5910
0.3	0.6179	0.6217	0.6255	0.6293
0.4	0.6554	0.6591	0.6628	0.6664
0.5	0.6915	0.6950	0.6985	0.7019

To determine the CDF, or probability that Z is less than 0.23, use the row for the first digit (0.2), and the column for the second digit (0.03):

$$P(Z < 0.23) = 0.5910 \tag{16}$$

A z-score tells us how unusual a measurement is relative to the population mean. The z-table converts this distance into a probability.

Module 1.2.2 : Probability Distribution Functions (PDFs)

Example : A missile system has a probability of failure $p_f = 0.07$ for each launch. Assume each launch is independent.

Question : What is the probability that at least 8 out of 10 missile launches are successful?

The binomial probability distribution is

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \tag{17}$$

Apply the formula with $n = 10$, and $p = 0.93$. For 8, 9, and 10 successful launches:

$$\begin{aligned} P(X \geq 8) = &\binom{10}{8}(0.93)^8(0.07)^2 \\ &+ \binom{10}{9}(0.93)^9(0.07)^1 \\ &+ \binom{10}{10}(0.93)^{1}0(0.07)^0 \\ &= 0.94 \end{aligned} \tag{18}$$

Result : The mission has a 94% success rate. The cumulative distribution function F for r or fewer successes in n trials may be written

$$F(r) = \sum_{x=0}^{r} \binom{n}{x} p^x q^{n-x} \tag{19}$$

Well-known PDFs can often be implemented in Microsoft Excel:

```
f(x) = BINOMDIST(x,n,p,FALSE)
F(r) = BINOMDIST(r,n,p,TRUE)
```

Example : A missile defense battery experiences an average of 2 missile guidance failures per year. Assume failures occur independently at a constant average rate.

Question : What is the probability that the battery experiences no more than one guidance failure during the next year?

The Poisson probability distribution is

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}. \quad (20)$$

Apply the formula with λ equal to 2 failures per year times one year.

$$P(X \leq 1) = P(0) + P(1) = 0.406 \quad (21)$$

Result : There is approximately a 40.6% probability that the battery experiences one or fewer guidance failures during the next year.

The cumulative distribution function (CDF) is

$$F(r) = \sum_{x=0}^{r} \frac{\lambda^x e^{-\lambda}}{x!}. \quad (22)$$

The PDFs in Microsoft Excel:

```
f(x)=POISSON(x,lambda,FALSE)
F(r)=POISSON(r,lambda,TRUE)
```

The Weibull distribution is widely used in reliability engineering to model the lifetime of mechanical components. The CDF is

$$F(t) = 1 - e^{-(t/\eta)^{\beta}} \tag{23}$$

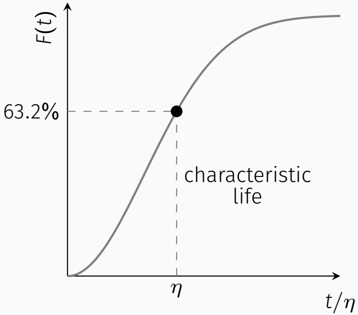
where:

- β = shape parameter
- η = characteristic life

Characteristic life:

$$F(\eta) = 1 - e^{-1} = 0.632 \tag{24}$$

Therefore, η is the time at which approximately 63.2% of components have failed.



Given Eq. **??**, prove that

$$R(t) = e^{-(t/\eta)^{\beta}} \tag{25}$$

$$f(t) = \frac{\beta t^{\beta-1}}{\eta^{\beta}} \exp\left[-\left(\frac{t}{\eta}\right)^{\beta}\right] \tag{26}$$

Example : A gearbox bearing has a lifetime modeled by a Weibull distribution. Based on historical test data, the bearing has: $\beta = 2.5$ and $\eta = 10,000$ hours.

Question : What is the probability that a randomly selected bearing survives beyond 8,000 operating hours?

The Weibull reliability function is

$$R(t) = P(T > t) = e^{-(t/\eta)^{\beta}} \tag{27}$$

Apply the values:

$$R(8000) = e^{-(8000/10000)^{2.5}} = 0.564 \tag{28}$$

Result : There is approximately a 56.4% probability that the bearing will survive beyond 8,000 hours of operation.

Excel functions : To implement the Weibull PDF/CDF in Microsoft Excel, use

```
CDF:
=WEIBULL.DIST(t,beta,eta,TRUE)
PDF:
=WEIBULL.DIST(t,beta,eta,FALSE)
```